Hybrid LLM + Higher-Order Quantum Approximate Optimization for CSA Collateral Management

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arXiv:2510.26217v1 [q-fin.CP] 30 Oct 2025

***Abstract*—We address *finance-native* collateral optimization under ISDA Credit Support Annexes (CSAs), where integer lots, Schedule A haircuts, RA/MTA gating, and issuer/currency/class caps create rugged, legally bounded search spaces. We introduce a certifiable hybrid pipeline *purpose-built* for this domain: (i) an evidence-gated LLM that extracts CSA terms to a normalized JSON (abstain-by-default, span-cited); (ii) a quantum-inspired explorer that interleaves simulated annealing with *micro higher-order QAOA* (HO-QAOA) on binding sub-QUBOs (subset size $n \leq 16$, order $k \leq 4$) to coordinate multi-asset moves across caps and RA-induced discreteness; (iii) a weighted risk-aware objective (Movement, CVaR, funding-priced overshoot) with an explicit coverage window $U \leq R_{\text{eff}} + B$; and (iv) CP-SAT as single arbiter to *certify* feasibility and gaps, including a U-cap pre-check that reports the minimal feasible buffer $B^\star$. Encoding caps/rounding as higher-order terms lets HO-QAOA target the domain couplings that defeat local swaps. On government bond datasets and multi-CSA inputs, the hybrid improves a strong classical baseline (BL-3) by 9.1%, 9.6%, and 10.7% across representative harnesses, delivering better cost–movement–tail frontiers under governance settings. We release governance-grade artifacts—span citations, valuation matrix audit, weight provenance, QUBO manifests, and CP-SAT traces—to make results auditable and reproducible.**

I. Introduction

Collateral posted under ISDA Credit Support Annexes (CSAs) must satisfy legally binding rules on eligibility, haircuts (Schedule A), rounding (RA), Minimum Transfer Amount (MTA), and concentration limits (issuer/currency/class/global). Integer lots, haircut tiers, and caps create a rugged search space; operational frictions (movement) and funding/tail considerations further complicate the objective. Enterprise diagnostics suggest that suboptimal allocation, trapped liquidity, and fragmented inventories impose material costs, motivating automation and enterprise optimization [1], [2], [3].

We present a *domain-specific, certifiable* hybrid pipeline for CSA-governed collateral allocation that integrates document understanding, higher-order discrete optimization, and formal certification:

1) *Evidence-gated CSA extraction.* An abstain-by-default LLM converts CSAs and related legal/financial documents into a normalized, CSA-aware JSON with span citations (thresholds, IA/IM, MTA, RA, eligibility and haircut matrices, regime selectors, caps, inventory metadata, scenarios).
2) *Hybrid explorer with micro higher-order QAOA (HO-QAOA).* We interleave quantum-inspired simulated annealing with *micro-HO-QAOA* on binding sub-QUBOs (subset size $n \leq 16$, interaction order $k \leq 4$), explicitly encoding rounding/caps as higher-order terms to coordinate multi-asset moves that defeat local swaps. This aligns with recent evidence that higher-order QAOA outperforms quadratic QAOA on rugged finance landscapes [10], [11]. We cap $k \leq 4$ to limit ancilla overhead and compilation depth.
3) *Weighted, risk-aware objective with funding-priced overshoot.* We scalarize operational and risk trade-offs as

$$\begin{aligned} J &= \text{BaseCost_abs} + \lambda\,\text{Movement} \\ &\quad + \mu\,\text{CVaR} + \gamma\left(U - R_{\text{eff}}\right)_+. \end{aligned} \tag{1}$$

   Here λ prices execution/ops churn, μ prices tail risk via CVaR, and γ prices funding on over-posted collateral ("overshoot") consistent with LVA/FVA [12], [13], [14], [15], [16], [17], [18]. We also enforce an explicit coverage window $U \leq R_{\text{eff}} + B$ to govern buffers.
4) *CP-SAT certification with feasibility diagnostics.* The incumbent is *certified* (status, bounds, gap) under identical constraints, and a U-cap pre-check reports the minimal feasible buffer $B^\star$ when windows are too tight.
5) *Governance-grade artifacts.* We emit span citations, a valuation matrix audit, weight-provenance JSON, QUBO manifests (subset n, order k, depth p), and CP-SAT traces (status, bounds, slacks) for auditability and reproducibility.

Upstream CSA-domain LLM. As an upstream stage, we train a CSA-domain LLM to extract key terms from CSAs and related documents (Schedules, Credit Support Deeds, eligibility matrices). The model is evidence-gated (abstain-by-default with span citations) and emits the CSA-aware data model that directly feeds the optimizer (see *CSA-Aware Data Model*). Full training data, model architecture, and benchmarks are covered in a separate paper.

Weighted scalarization and provenance. Our weighted formulation traces Pareto-efficient trade-offs [12], [13], with CVaR capturing tail exposure [14], movement reflecting execution frictions [15], and γ dailyizing funding spreads per LVA/FVA principles [16], [17], [18]. We calibrate (λ, μ, γ) from observed ops costs, tail pricing, and funding bps, and record inputs/units in a weights-provenance artifact for governance.

Positioning and comparisons. By targeting higher-order domain couplings (RA/MTA interactions and concentration caps) with micro-HO-QAOA, and certifying outcomes with CP-

SAT, our pipeline improves cost–movement–tail frontiers on realistic government bond datasets and multi-CSA inputs.

II. Background and Related Work

A. Collateral Optimization

Classical formulations encode haircut schedules, eligibility, and concentration limits, with rounding to RA and MTA gating. Pricing practice introduces liquidity/funding adjustments: the Liquidity Valuation Adjustment (LVA) discounts cash collateral at rate r_c vs. risk-free r; FVA reflects funding costs on uncollateralized parts [1]. Operating models emphasize enterprise views and the six levers—Documentation, Automation, Transformation, Optimization, Mobilization, Segregation—[3]. We retain MILP/CP-SAT certification and augment exploration with quantum-inspired sampling and micro-HO-QAOA near binding corners, shaping the objective with movement penalties and Weighted-CVaR.

B. Related Work

LLMs for CSA extraction. Evidence-gated LLMs achieve $90\%+$ clause-level accuracy for thresholds, MTA, eligibility, and haircut schedules mapped to CDM-like schemas [4].

Collateral & liquidity efficiency. Guidance urges minimizing trapped liquidity, balancing movement, and reserving buffers [5].

Quantum(-inspired) optimization. QUBO mappings and NISQ-era methods motivate micro-QUBOs near binding constraints [6], [7].

Hybrid solvers. QAOA/VQE sampling paired with classical local search improves quality under resource limits [8]. Hardware performance milestones suggest headroom for small structured QAOA in workflows [9].

Higher-order QAOA for finance. Closest to our setting, Uotila, Ripatti, and Zhao extend QAOA to *higher-order* (HUBO) portfolio optimization and report 15–25% gains over vanilla (quadratic) QAOA on rugged financial landscapes for n=8–24 variables on NISQ simulators [10], [11]. Their formulation explicitly models multi-asset interactions (e.g., covariance/risk and cardinality) as $k>2$ terms and uses *order-aware partitioning* and spectral grouping to set subset sizes n (base n=8–12 for k=2, add 4–8 for constraints). We borrow three elements: (i) treating CSA caps/eligibility and MTA/rounding couplings as higher-order penalties in micro-HO-QAOA (e.g., using k=3 terms to model window/MTA interactions and multi-cap couplings); (ii) selecting $n\approx 8$–16 via spectral clustering of highly coupled lots, which aligns with their n recommendations and our ancilla budget; and (iii) warm-starting quantum jumps from a classical incumbent (our CP-SAT/SA incumbent), which their results show mitigates barren plateaus. Conceptually, their "integer shares" mirror our discrete lots x_i, and their eligibility screens map to our CSA-based haircut/eligibility flags, making their method particularly applicable to ISDA-CSA collateral allocation.

Benchmarking and noisy regimes. Recent studies benchmark QAOA/HO-QAOA and related hybrids for finance portfolios in noisy settings, including VQE-style variants and noise-aware compilations (add exact citations). We position our *micro*-HO-QAOA as a targeted jump operator embedded in a certified pipeline rather than a stand-alone solver, and we cap $k\leq 4$ to control ancilla overhead.

III. Problem Formulation

We pick integer lots $x_i \in \mathbb{Z}_{\geq 0}$ for eligible assets i with after-haircut value v_i and daily carry cost c_i. Coverage $U = \sum_i v_i x_i$. The effective requirement uses RA rounding:

$$R_{\text{eff}} = \left\lceil \frac{\max(E-T-\text{IA}-\text{IM},0)}{\text{RA}} \right\rceil \text{RA}. \tag{2}$$

We enforce $U \geq R_{\text{eff}}$, an optional cap $U \leq R_{\text{eff}}+B$, and cash/issuer/class/currency/global caps.

a) Objective.:

$$\begin{aligned} \min\ J \ =\ & \sum_i c_i x_i \ +\ \lambda \|x-h\|_1 \\ & +\ \mu\, \text{CVaR}_(Lx) \ +\ \gamma\,(U-R_{\text{eff}})_+. \end{aligned} \tag{3}$$

CVaR uses a linearization (τ, z_s) with scenario weights $\sum_s w_s = 1$.

b) Binary/QUBO view.: Integer lots are encoded via bounded binaries $y_{i\ell} \in \{0,1\}$ s.t. $x_i = \sum_{\ell=1}^{m_i} y_{i\ell}$ with per-lot valuation $v_{i\ell}=v_i$ and costs $c_{i\ell}=c_i$.

c) HO-QAOA definition and (n,k) roles.: We construct a higher-order Ising Hamiltonian

$$H_P \ =\ \sum_j a_j Z_j + \sum_{j<k} b_{jk} Z_j Z_k + \sum_{j<k<\ell} c_{jk\ell} Z_j Z_k Z_\ell + \cdots$$

where higher-order ($k \geq 3$) terms encode multi-asset interactions from caps (issuer/class/currency/global), window coupling (U near R_{eff}), and lot granularity. The *order* k denotes the maximum Pauli-Z tensor product degree needed to represent constraints/objective couplings in the subproblem. We use a *micro-HO-QAOA* on *subsets* of variables of size n (typically 8–16) selected near binding corners. The HO-QAOA state of depth p is

$$|\gamma,\beta\rangle \ =\ \prod_{\ell=1}^{p} \left(e^{-i\beta_\ell \sum_j X_j}\ e^{-i\gamma_\ell H_P} \right) |+\rangle^{\otimes n},$$

with standard X-mixer; higher-order phase operators $e^{-i\gamma_\ell Z_{j_1}\cdots Z_{j_k}}$ are compiled either directly or via ancillas. For $k > 2$, ancilla qubits linearize/multiply higher moments; if ancillas inflate the subset above $n_{\max}$, we skip the quantum jump for that iteration and log the reason.

Impact of n. Larger n captures more coupled moves across caps/rounding but increases circuit width and optimizer complexity; empirically, $n \in [8,16]$ balances expressivity and run time, reliably crossing rugged neighborhoods that defeat local swaps. *Order/size crosswalk.* Our practical caps ($k\leq 4$) and subset limits ($n \leq 16$) follow the order-aware guidance observed in higher-order finance QAOA benchmarks, which report best empirical trade-offs around $n\approx 12$–18 for $k>2$ on rugged landscapes with warm starts [10], [11].

Impact of k. Higher k allows direct encoding of multi-way caps and overshoot couplings; however, gate compilation depth and noise rise with k. We cap at $k\leq 4$ in practice; above this, we fall back to classical exploration.

IV. CSA-Aware Data Model

As an upstream stage, we train a CSA-domain LLM to extract key terms from CSAs and related financial/legal documents (e.g., Schedules, Credit Support Deeds, annexed eligibility matrices). The model is evidence-gated (abstain-by-default with span citations) and emits a normalized, CSA-aware data model that includes terms (Threshold, IA/IM, MTA, RA), eligibility and haircut matrices, regime selectors, concentration caps, inventory metadata, and scenario inputs.We standardize those extraction parameters as inputs/outputs in a governance-ready JSON schema. Key fields:

A. Counterparty & Legal

- `csa.meta`: governing law (NY/English), bilateral/one-way.
- `csa.terms`: Threshold T, Independent Amount IA, Initial Margin IM, Minimum Transfer Amount (MTA), Rounding Amount (RA), Base Currency, FX conventions.
- `csa.regime`: valuation regime selector in Schedule A; the default may be overridden per asset bucket.
  - `sp`: S&P column (`sp_pct`)
  - `m1`: Moody's First (`m1_pct`)
  - `m2`: Moody's Second (`m2_pct`)

B. Valuation Haircuts and Eligibility

- `haircuts.matrix`: haircut percentage indexed by (ICAD, bucket, regime).
- `eligibility.scheduleA`: eligible asset classes and buckets (Govt, Agency, Corp, MBS, TIPS, Cash), issuer ratings/tenor constraints.

C. Caps and Windows

- `caps`: `cash_cap` (e.g., 20% of U), `issuer_cap`, `class_cap`, `currency_cap`, `global_cap`.
- `window`: policy buffer B (bps or \$), optional hard coverage cap $U \leq R_{\text{eff}} + B$.

D. Exposure and Scenarios

- `exposure`: E (base currency) and timestamp; optional path of E_t for rolling re-optimization.
- `scenarios`: matrix L (per-asset loss/PNL across scenarios) with weights w_s (normalized for CVaR).

E. Inventory and Costs

- `inventory`: items with `id`, `class`, `issuer`, `bucket`, `currency`, price, unit, current lots h_i, and per-lot valuation v_i after haircut.
- `costs`: daily carry c_i (\$/lot/day), operational move cost unit for movement.

F. Weights and Provenance

- `weights`: (λ, μ, γ) with calibration inputs and units: λ (ops amortization per lot over horizon), μ (price per \$MM CVaR per day), γ (funding bps $\rightarrow$ daily carry).
- `weights_provenance`: calibration inputs (ops move cost, horizon days, CVaR price, funding bps), hash, and timestamp.

Algorithm 1 Micro-HO-QAOA Jump (Explore)

In: incumbent x, objective J, graph G, limits $(n_{\max}, k_{\max}, p)$, plateau (S, ϵ), optional angles $(\gamma^{(0)}_{1:p}, \beta^{(0)}_{1:p})$

if $\textsc{Plateau}(x, S, \epsilon) = \textsc{false}$ **then**
  return x
$S \leftarrow \textsc{SpectralSelect}(G, n_{\max})$ ▷ *$|S| \leq n_{\max}$ (typ. 8–16)*
$H_P \leftarrow \textsc{BuildHubo}(S, k_{\max})$ ▷ *RA/MTA, window, caps; ancillas for $k>2$*
$w \leftarrow \textsc{AncillaWidth}(H_P)$
if $w > n_{\max}$ **then**
  return x ▷ *skip jump; continue SA + repair*
$|\psi| \leftarrow \text{PREP}(w)$; optionally $(\gamma_\ell, \beta_\ell) \leftarrow (\gamma^{(0)}_\ell, \beta^{(0)}_\ell)$
for $\ell = 1$ **to** p **do** ▷ *mixer ramp allowed*
  $|\psi| \leftarrow \text{MIX}(\beta_\ell)\big(\text{PHASE}(\gamma_\ell, H_P)(|\psi|)\big)$
$z \sim |\psi|$; $y \leftarrow \textsc{MapLots}(z)$ ▷ *ancillas→vars*
$y \leftarrow \textsc{Repair}(y)$ ▷ *caps/RA/MTA/window*
$\tilde{x} \leftarrow x$; $\tilde{x}_S \leftarrow y_S$
if $\textsc{Feasible}(\tilde{x})$ **and** $J(\tilde{x}) < J(x)$ **then**
  return $\tilde{x}$ ▷ *accept*
else
  return x ▷ *reject*

G. Governance/Audit Toggles

- `audit.flags`: enable span citations, valuation audit, QUBO manifests, CP-SAT traces.
- `solver.limits`: SA iterations, HO-QAOA $n_{\max}$, $k_{\max}$, depth p, and wall constraints.

V. Hybrid Pipeline (Explore $\rightarrow$ Prove $\rightarrow$ Explain/Audit)

We create the full workflow with four phases:

A. Phase 1: Explore (Search)

1) **Initialization:** Compute R_{eff} via (2); derive per-lot v_i from haircuts; seed with BL-1 (density greedy).
2) **Local search:** Simulated annealing (integer neighborhoods; add/swap/remove) with feasibility repair (caps, RA, MTA, window).
3) **Spectral subset selection:** Build a unitless interaction graph (dual/gradient proxies, feasibility slacks) and pick top-K nodes by $|\text{dual}|$; prune edges by ε to stabilize.
4) **Micro-HO-QAOA jump:** If improvement $<0.3\%$ over S SA steps, form a sub-QUBO on $n \leq 16$ variables (with ancillas if $k > 2$) and perform one HO-QAOA jump (depth p); accept if J decreases and feasibility holds; otherwise revert.Please see the Algorith 1: Micro-HO-QAOA Jump (Explore).

B. Phase 2: Prove (Certification)

We pass the incumbent to CP-SAT with the same constraints and objective components (linearized CVaR and overshoot). We report: status (OPTIMAL/FEASIBLE/INFEASIBLE), incumbent/best bound, MIP gap, and per-constraint slacks.

C. Phase 3/4: Explain & Audit (Governance)

We emit governance HTML with: objective breakdown; valuation matrix audit; weight provenance; spectral/QUBO manifests (subsets, n, k, p); and CP-SAT traces (status, bounds, slacks). Reproducibility hashes and seeds are included.

D. Baselines and Feasibility (Overshoot & $B^\star$)

a) Baselines.: We benchmark three progressively stronger heuristics:

- **BL-1 (density greedy, cap-safe):** ranks assets by cost-to-valuation density and fills to the window under caps; fast, but can stall near binding corners.
- **BL-2 (bucket-first greedy + repair):** prioritizes bucket/cap compliance during greedy fill, then repairs to align with the window; tighter coverage, typically higher movement.
- **BL-3 (BL-1 seed + 2-opt swaps):** starts from BL-1 and applies local pairwise swaps to reduce cost while respecting feasibility; strong local polish, but prone to plateaus.

Hybrid. Uses BL-3 as a seed, then interleaves simulated annealing with a spectral micro-HO-QAOA jump to cross binding constraints and escape BL-3 plateaus, followed by local repair for feasibility.

b) Overshoot and Feasibility: Because lots are discrete, $U{=}R_{\text{eff}}$ is rare. We compute a minimal feasible buffer $B^\star$ by (i) building any feasible cover without the U-cap, then (ii) greedily reducing U while preserving caps/RA. If the user-specified buffer $B < B^\star$, we flag `infeasible_u_cap` and report $B^\star$ (USD and bps). The objective's overshoot penalty $\gamma\left(U - R_{\text{eff}}\right)_+$ trades off carry versus buffer.

VI. Case Study

A. CSA Summary

Governing law. 2009 New York–law CSA, bilateral.
Base currency & eligibility. USD base; USD/EUR cash and securities per Schedule A (government, agencies, corporates, TIPS, MBS), valuation by rating/tenor.
Threshold/MTA/Rounding. $T = 0$, $\text{IA} = 0$, $\text{IM} = 0$; MTA $= \$100{,}000$; RA $= \$10{,}000$.
Valuation regime. Moody's First (m1) default; S&P (sp) and Moody's Second (m2) available.
Operational caps. Buffer $B{=}25$ bps of R_{eff}; cash cap $= 20\%$ of U.
Exposure. $E = \$130{,}340{,}000$; R_{eff} computed via (2).
Inventory proxy. USD cash and UST ladder (6M–20Y), TIPS, Agency, AAA MBS, IG Corps; per-lot v_i after haircuts; lots aligned to RA (cash) and \$1MM coupons (bonds).

B. Valuation Regimes

We consider **sp**, **m1** (default), and **m2**, using the Schedule A matrix for haircuts.

C. Objective and Constraints (shared)

Minimize $J = \text{BaseCost_abs} + \lambda\,\text{Movement} + \mu\,\text{CVaR} + \gamma\,\text{Overshoot}$, subject to $R_{\text{eff}} \le U \le R_{\text{eff}}{+}B$, cash cap, and integer-lot availability. Units: BaseCost_abs [\$/day], Movement [lots], CVaR and Overshoot [\$].

D. How to Choose Weights (practical guidance)

We calibrate (λ, μ, γ) from operational inputs: (i) per-lot Ops move cost and amortization horizon $\Rightarrow \lambda$, (ii) daily price for 1\$MM CVaR $\Rightarrow \mu$, (iii) annual funding bps $\Rightarrow \gamma$ via day-count. Each run logs a `weights_provenance.json` (inputs, units, calibrated triplet, hash).

E. CP-SAT Results and Meaning

CP-SAT returns `OPTIMAL` when the incumbent attains the global minimum and the MIP gap is zero; `FEASIBLE` when a feasible incumbent exists with a nonzero bound-gap; `INFEASIBLE` when no solution satisfies caps/window/RA. For each case we report per-constraint slacks (cash/issuer/-class/currency/global), confirming which limits bind.

F. Harness Setups and Results

We analyze three scenarios. Across harnesses A/B/C, the Hybrid improves the BL-3 objective by **9.1%**, **9.6%**, and **10.7%**, respectively (see tables below).

- Units: *BaseCost_abs* [\$/day], *Movement* [lots], *CVaR*$_{0.90}$/*Overshoot*/*UsedValue* [\$]; all rounded to 2 dp.
- CVaR weights normalized ($\sum w = 1.0$); governance HTML warns if renormalization occurred.
- Weight provenance (`.json`) and valuation audit are linked in each governance HTML.

a) Harness A: m1, buffer 25 bps, cash cap 20%, practical weights.:

Model	BaseCost	Movement	CVaR	Overshoot	J
BL-1	100.0	28	540,000	210,000	1,12x
BL-2	99.1	35	528,000	195,000	1,10x
BL-3	98.7	24	520,000	182,000	1,00x
Hybrid	98.4	22	515,000	155,000	**0.91x**

- **Configuration:** m1 regime; buffer $B = 0.25\%$ of R_{eff}; cash cap $= 20\%$ of U_{cap}; weights $\approx (\lambda, \mu, \gamma) = (30.0,\ 0.001,\ 1.39 \times 10^{-5}\ \text{day}^{-1})$.
- **Intent:** "Everyday" governance settings with moderate funding and moderate tail price; tests balanced trade-offs.
- **Effect:**
  - γ penalizes overshoot enough to cut excess usage without exploding Movement.
  - μ applies light tail pressure; λ moderates lot churn.
  - Subset size n stays $\approx$ 8–16; must-jump triggers rarely.
- **Result (vs BL-3):** Hybrid improves Objective by $\approx 9.1\%$ with lower *Movement* and *Overshoot*; breakdown shows most gains from $\gamma{\cdot}$Overshoot, with some from $\mu{\cdot}$CVaR.

Conclusion: Hybrid reduces J by $\approx 9.1\%$ vs BL-3, primarily by trimming Overshoot at similar BaseCost and slightly lower Movement.

b) Harness B: m1, buffer 10 bps, cash cap 15%, tight-liquidity weights (higher γ).:

Model	BaseCost	Movement	CVaR	Overshoot	J
BL-1	101.3	31	556,000	132,000	1,14x
BL-2	100.6	37	544,000	121,000	1,11x
BL-3	100.2	25	536,000	113,000	1,00x
Hybrid	100.0	24	533,000	91,000	**0.904x**

- **Configuration:** m1; buffer $B = 0.10\%$; cash cap $= 15\%$; weights $\approx (\lambda, \mu, \gamma) = (28.57,\ 0.0025,\ 2.22 \times 10^{-5}\ \text{day}^{-1})$.
- **Intent:** Tighter liquidity and higher funding pressure; tests robustness when overshoot is expensive and buffer small.
- **Effect:**
  - Larger γ materially suppresses overshoot, trading some BaseCost/Movement.
  - Higher μ drives tail reduction; λ still curbs churn.
  - $n \approx 8$–16; must-jump fires more often to escape SA plateaus.
- **Result (vs BL-3):** Hybrid improves Objective by $\approx 9.6\%$; gains mainly from γ·Overshoot and μ·CVaR, with Movement contained by λ.

Conclusion: With tighter buffer and cash cap, overshoot control dominates. The must-jump rule breaks SA plateaus; J improves $\approx 9.6\%$ vs BL-3.

c) Harness C: m2, buffer 25 bps, cash cap 20%, practical weights.:

Model	BaseCost	Movement	CVaR	Overshoot	J
BL-1	99.5	27	501,000	204,000	1,13x
BL-2	99.0	33	492,000	193,000	1,08x
BL-3	98.6	23	485,000	178,000	1,00x
Hybrid	98.3	22	480,000	149,000	**0.893x**

- **Configuration:** m2; buffer $B = 0.25\%$; cash cap $= 20\%$; weights $\approx (\lambda, \mu, \gamma) = (30.0,\ 0.001,\ 1.39 \times 10^{-5}\ \text{day}^{-1})$.
- **Intent:** Regime sensitivity with tighter haircuts; tests ability to coordinate under higher required usage/tail.
- **Effect:**
  - Tighter valuations raise UsedValue and CVaR; Hybrid's spectral $n \approx 8$–16 helps cross binding corners (window/caps/lot granularity).
  - Must-jump occasionally assists when caps bind.
- **Result (vs BL-3):** Hybrid improves Objective by $\approx 10.7\%$; breakdown shows meaningful $\gamma \cdot$Overshoot and μ·CVaR reductions while keeping Movement controlled.

Conclusion: Under tighter m2 haircuts, Hybrid improves J by $\approx 10.7\%$ vs BL-3, keeping n within 8–16 via spectral capping.

G. Weight Selection: Why These Numbers

We target business trade-offs: (i) if Ops capacity is constrained, increase λ to suppress Movement; (ii) if funding costs dominate, raise γ to push $U \downarrow$ (less overshoot); (iii) if tail discipline is paramount, raise μ (CVaR), accept modest BaseCost/Motion increases. Calibration is documented in the weight provenance blob and mirrored in governance HTML.

VII. Governance

We produce:

- **Span citations (LLM extraction):** prompt hash and source spans for each clause (Threshold, MTA, RA, eligibility, haircuts).
- **Valuation matrix audit:** table mapping *instrument* $\rightarrow$ *ICAD/bucket/regime* $\rightarrow$ *haircut%* $\rightarrow v_i$ for full reproducibility.
- **Weight provenance:** calibration inputs/units and (λ, μ, γ), with hashes/timestamps.
- **QUBO manifests:** for each jump: subset IDs, n, k, p, compiled terms, and acceptance decision.
- **CP-SAT traces:** status (OPTIMAL/FEASIBLE/INFEASIBLE), incumbent, best bound, gap, and per-constraint slacks (cash/issuer/class/currency/global); infeasible windows include $B^\star$.

VIII. Ablations

Spectral stability. Bounding edge weights to $[0, 1]$ and ε-pruning yield stable cluster selection; without pruning, acceptance variance rises.

Subset size n and cap. Performance saturates around $n \approx 12$; $n<8$ underfits multi-way caps; $n>16$ adds overhead and ancilla pressure with diminishing returns. We hard-cap $n \leq 16$.

Order k and ancillas. Enabling $k = 3$ captures issuer/class/currency triples and RA/MTA window couplings; $k=4$ further improves near tight windows at higher compilation cost. We cap $k \leq 4$ to contain ancilla-expanded width.

γ/μ sweeps. Increasing γ drives overshoot $\downarrow$ and BaseCost $\uparrow$ monotonically; increasing μ reduces tail exposure with modest Movement increase. Hybrid dominates BL-3 along both trade-off frontiers.

Must-jump rule. Enforcing at least one HO-QAOA jump after S low-improvement SA steps avoids long plateaus and drives consistent Overshoot reductions.

Weighted-CVaR. Pricing tails ($\mu > 0$) smooths the BaseCost–Overshoot frontier and reduces solution churn across scenario sets; renormalization warnings are emitted if $\sum_s w_s \neq 1$ and auto-fixed.

Overshoot penalty γ. Sweeps show monotone Overshoot$\downarrow$ and BaseCost$\uparrow$; Hybrid dominates BL-3 along this frontier, indicating effective cross-cap coordination.

Fallback behavior. If ancillas inflate n past $n_{\max}$ or compilation fails, the iteration logs a skip and reverts to SA/BL-3 neighborhoods; solution quality degrades gracefully.

IX. Conclusion

We presented a domain-specific, certifiable hybrid optimizer for CSA-governed collateral that unifies evidence-

gated CSA extraction, quantum-inspired search, and higher-order QAOA micro-jumps with CP-SAT certification. By encoding RA/MTA interactions and concentration limits as higher-order couplings, our method coordinates discrete lot moves that defeat purely local heuristics, while a weighted objective (movement, CVaR, funding-priced overshoot) captures the operational and risk economics of posting. Across realistic government bond datasets and multi-CSA inputs, the pipeline—*extract, explore, certify, audit*—consistently improves cost–movement–tail frontiers over strong classical baselines and yields governance-grade artifacts suitable for operational sign-off.

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